

Complementary Feeding — Highlighted Key Points

Indian Academy of Pediatrics (IAP) Guidelines

Q1 — What is Complementary Feeding?

A food containing proteins, carbohydrates, minerals, fats, vitamins, fibers is 'Complete Food'; the one with these in appropriate proportions is 'Balanced Food'.

Examples of balanced foods:

- Cereal (rice/wheat) + pulses/nuts + vegetables cooked in oil/ghee = balanced food (e.g. Upma, Pulao, Biryani, Poha).
- Cereals + nuts/raisins added to milk = balanced (e.g. Dalia, Kheer).

Fig. 1: Combination of these makes a balanced food

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Pyramid Level	Food Group	Examples
Tip (use sparingly)	Fat / Oil, Salt, and Sugar	Oil, ghee, sugar, salt
Second tier	Milk & Alternatives Meat, Fish, Egg & Alternatives	Milk, curd, paneer, eggs, chicken, fish
Third tier	Vegetables Fruits	Carrot, spinach, tomato, mango, banana, papaya
Base (eat most)	Grains, Roots, and Tubers	Rice, wheat, ragi, jowar, potato

A balanced diet combines foods from ALL levels of the pyramid. No single food is complete — variety is key!

Q2 — When to Start & Why?

The other reasons for complementary feeding at 6 months of age are:

- Child develops neck/head control and hand-to-mouth coordination.
- Child starts enjoying mouthing and biting.
- The intestines are mature and ready to digest pulses and cereals.
- Baby likes chewing and gumming semisolids as there is hardening of gums and eruption of teeth.
- Tendency to push solids out of mouth decreases.

Q3 — Which Foods to Use?

Appropriate food consistency: The food that just drips slowly from spoon and not flows.

- Every community has a staple food — the food that forms the main bulk; for example, wheat, rice.
- Staple foods can be cooked, served, and are good sources of energy and protein.

TABLE 2 — Feeding Schedule:

TABLE 2: Quantity, Frequency & Texture of Complementary Foods

Amounts of food to offer

Age	Texture	Frequency	Amount/meal
6–8 months	Mashed foods or thick porridges	2–3 meals/day + frequent breastfeeding	2–3 tablespoons
9–11 months	Mashed, finely chopped, or finger foods	3 meals + 1–2 snacks + breastfeeding	½ cup / bowl (125 mL)
1–2 years	Staple family foods, mashed/chopped if needed	3–4 meals + 1–2 snacks + breastfeeding	¾–1 cup / bowl (250 mL)

Note: For non-breastfed children — give additional 1–2 cups of milk + 1–2 extra meals daily.

→ How to modify nutritive value (highlighted):

- Nutritive value can be improved by increasing energy density by addition of ghee, sugar, jaggery, vegetable oils, butter, etc.
- Protein content can be improved by combining cereals and pulses to ensure delivery of all essential proteins; also by milling, germination, and fermentation.
- Viscosity can be reduced by malting so the child can eat more. Malted cereal/pulse provides more energy. Amylase-rich flour (ARF) increases energy density.

→ Vegetables & Fruits (highlighted):

- Feed with thick but smooth mixtures. A young infant around 6–9 months requires thick and smooth food preparations.
- Fortification with micronutrients: iodine, zinc, iron, vitamin D, vitamin A, calcium, phosphorus, etc.
- Green leafy vegetables, carrots, pumpkin, and seasonal fruits (papaya, mango, chikoo, banana) are important sources of vitamin A, B, and iron.

Q4 — Dietary Diversity (TABLE 4)

TABLE 4: The Various Food Groups	
Recommended Food Group	Examples
Grains, roots, and tubers	Rice, wheat, maize, jowar, ragi, potato, sweet potato
Legumes and nuts	Pulses, nuts, oil-seeds, dry fruits
Vitamin-A rich fruits & vegetables	Mango, carrot, papaya, tomato (orange/yellow/green)
Other fruits & vegetables	Locally available, fresh, seasonal, inexpensive
Dairy products	Milk, curd, yogurt, butter, paneer
Eggs	Eggs
Flesh foods	Meat, fish, poultry, organ meats

Q5 — Role of Feeding in Child's Growth & Development

- Parents should promote direct feeding and assist children in self-feeding.
- Many food preparations may be refused; try different food combinations, tastes, textures, and methods of encouragement.

- Feeding times are periods of learning and love — talk to children while feeding; maintain eye-to-eye contact.
- Avoid comparison with other children or siblings regarding choice of food.
- Children under 1 year should be fed by mother/caregiver. Toddlers fed in separate plate promoting self-feeding. Eating together as family improves appetite and avoids distractions.
- Do not force-feed the child. This way the child starts disliking food.
- Distractions such as viewing TV or mobile phone while feeding is strongly discouraged.

Q6 — Role of Commercial Foods

Many commercial foods are expensive and often have exaggerated health claims. Packaged food may not be a healthier or suitable option. Child's food should be prepared at home using commonly available ingredients. Commercial foods for infants and young children (<2 years) should not be promoted.

Q9 — Can Some Foods Cause Allergy?

Some babies may be allergic to cow's milk protein or soya protein. Others may react to eggs, peanuts, food additives, and spices. These allergies are not very common in Indian children. Consult your pediatrician if you feel your child is allergic to a particular food.

Q10 — Myths & Misconceptions (Highlighted)

- Locally available cheap seasonal fruits are NOT inferior. Costly/imported/cold-stored/tinned fruits or preserved juices may contain chemicals — avoid.
- Health drinks, supplements, and protein powders with 'exaggerated claims' are NOT more nutritious than home food.
- Any products promoted by celebrities may not be good. Biased and misguided advertisements mislead even educated individuals.
- Ultra-processed, attractively packed, cold-storage preserved foods are NOT 'magic foods'.
- Artificially prepared, flavored, and packaged commercial foods are NOT innovative or ideal for children.

BOX 1: Principles to Follow for Complementary Feeding

- Start complementary feeds after completion of 180 days (6 months).
- Continue breastfeeding up to 2 years of age with appropriate complementary feeding.
- Give semi-solid food. Avoid watery food such as soups, fruit juices, and animal milk.
- Prefer home-made food — clean, fresh, cheap, and easily available.
- Prefer balanced food (Cereal + Pulse + Vegetables) using locally available ingredients.
- Introduce one food at a time; once the child accepts it, introduce another.
- Add ghee, oil, oil-seed powder, fats to increase energy and palatability (except if overweight/obese).
- Respect child's choice and preferences. Avoid force feeding.
- Give as much as child consumes. Monitor weight regularly.

BOX 2: Things to Avoid in Complementary Feeding

- Avoid delay in starting complementary feeding.

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| • Avoid outside, artificial, packaged, commercial, and junk foods. |
| • Avoid foods with excess sugar, salt, and trans-fatty acids. |
| • Avoid ultra-processed and refined foods. |
| • Bottle-feeding has only disadvantages — avoid it. |
| • Avoid feeding while watching television or mobile phones. |
| • Do not force feed. Feeding should not be an unpleasant experience. |
| • Avoid overfeeding. |
| • Avoid foods that may cause choking (whole nuts, grapes, raw carrots). |

Guiding Principles for Complementary Feeding

- By 8 months, babies can eat 'finger foods' – foods that they hold themselves
- Increase the number of times food is offered.
- Offer 2 to 3 times a day at 6-8 months
- 3 to 4 times a day at 9-24 months, plus nutritious snacks* once or twice a day, as desired, in addition to breastfeeding

**Healthy food eaten between meals*

Amounts of Food to Offer

AGE	TEXTURE	FREQUENCY	AMOUNT OF EACH MEAL	% CALORIE FROM BREASTMILK	CALORIES FROM COMPLEMENTARY FEEDING	BREAST MILK SUBSTITUTES
0-6 months		> 8 times per day	-	100%	-	150 ml/kg
At 6 months	Soft Porridge, well mashed vegetable, meat fruit	2 times per day plus frequent breastfeeds	2-3 tablespoonful	-	-	-
7-8 months	Mashed foods	3 times per day plus frequent breastfeeds	Increasing gradually to 2/3 of a 250 ml cup or more than ¾ of a 200 ml cup at each meal	70%	200 kcal	600 ml/day
9-11 months	Finely chopped or mashed foods, and foods that baby can pick-up	3 meals plus 1 snack between meals plus breastfeeds	¾ of a 250 ml cup or a full 200 ml cup	55%	300 kcal	550 ml/day
12-24 months	Family foods, chopped or mashed if necessary	3 meals plus 2 snacks between meals plus breastfeeds	A full 250 ml cup/bowl or more than a full 200 ml cup/bowl	40%	550 kcal	500 ml/day

Food Square 3 G's

Grains/Cereals/Staple Rice/Wheat/Ragi/Bajra. Provides carbohydrates 55%		Grams/Dals/Pulses/Nuts All dals and grams Thor dal/green dhal/black dhal. Groundnuts/meat/poultry/fish. Provides proteins. 20%
	Exclusive Breastfeeding for 6 months and continued breastfeeding up to 2 years and beyond	
Greens/vegetables/fruits (greens, yellow, orange, and red) Any amount. Provide micronutrients/vitamin A/B complex/vitamin C/fiber/trace elements.	Sugar/Jaggary/oil = empty calories	Fats and oils. Ghee/butter/oil/nuts. Provides fat soluble vitamins and taste. 30%

Stomach Capacity is 30ml/KG

Fiber is Now Being Considered as the New Macro Nutrient

- 1 bowl (200ml) of fruits = 6 grams of fiber.
- Rule of thumb: age in years + 25 to 30 grams of fiber per day.

Fruit Juices

- Encourage intake of regional and seasonal whole fruits over fruit juices in children and adolescents. Fruit juices/fruit drinks/SSBs should not be offered to infants and young children aged below 2 years.
- For children and adolescents (2-18 years) fruit juices, fruit drinks and SSBs should be avoided as far as possible.

- Water should be encouraged as the best drink and should be promoted over fruit juices/drinks at home and school.
- Fruit juices/drinks, if given, should be limited to 125 mL per day for children aged between 2-5 years, and 250 mL per day for age >5 years; and these should preferably be given as fresh juices.